

Exercise 4.1.19

$$y(t) = c_1 \cos \omega t + c_2 \sin \omega t$$

$$\begin{bmatrix} c_1 & c_2 \end{bmatrix} \begin{bmatrix} \cos \omega t \\ \sin \omega t \end{bmatrix} = c_1 \cos \omega t + c_2 \sin \omega t = y(t)$$

$$f(t) = a_1 \cos \omega t + a_2 \sin \omega t$$

$$g(t) = b_1 \cos \omega t + b_2 \sin \omega t$$

Closure under addition

$$\begin{aligned} f(t) + g(t) &= a_1 \cos \omega t + a_2 \sin \omega t + b_1 \cos \omega t + b_2 \sin \omega t \\ &= (a_1 + b_1) \cos \omega t + (a_2 + b_2) \sin \omega t \end{aligned}$$

$$a_1, b_1, a_2, b_2 \in \mathbb{R} \quad (a_1 + b_1), (a_2 + b_2) \in \mathbb{R}$$

Closure under scalar multiplication

$$\begin{aligned} c \cdot f(t) &= c(a_1 \cos \omega t + a_2 \sin \omega t) \\ &= ca_1 \cos \omega t + ca_2 \sin \omega t \end{aligned}$$

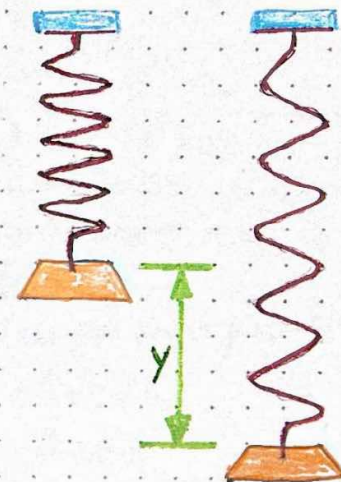
$$a_1, a_2, c \in \mathbb{R} \quad ca_1, ca_2 \in \mathbb{R}$$

Zero vector

Here, zero vector means $f(t) = 0$ for any t .

By selecting $c_1 = c_2 = 0$, we get the zero vector.

$$V = \text{span}\{\cos(\omega t), \sin(\omega t)\}$$



Axioms

1. $\vec{u} + \vec{v}$ is in V
2. $\vec{u} + \vec{v} = \vec{v} + \vec{u}$
3. $(\vec{u} + \vec{v}) + \vec{w} = \vec{u} + (\vec{v} + \vec{w})$
4. There is a zero vector
5. For each $\vec{u}$ in V , there is a $-\vec{u}$ such that $\vec{u} + (-\vec{u}) = \vec{0}$
6. $c\vec{u}$ is in V (scalar multiplication)
7. $c(\vec{u} + \vec{v}) = c\vec{u} + c\vec{v}$
8. $(c+d)\vec{u} = c\vec{u} + d\vec{u}$
9. $c(d\vec{u}) = (cd)\vec{u}$
10. $1\vec{u} = \vec{u}$